OWNER	:	**INDIAN OIL CORPORATION LIMITED (IOCL)**
PMC	:	**TOYO ENGINEERING INDIA PRIVATE LIMITED (TEIPL)**
LSTK CONTRACTOR	:	**L&T HYDROCARBON ENGINEERING LIMITED (LTHE)**
UNIT / UNIT NO	:	**DOCK LEVELLER FOR TEG WELDING AREA / 207D**
PMC JOB NO. / NAME	:	**6389 / IPEG**
LTHE JOB NO.	:	**IOCL-O&U**
VENDOR	:	**NIHVA TECHNOLOGIES PVT LTD**
PO NUMBER	:	**Petrochemicals/75000-65110/**
ITEM NAME & TAG NO.	:	**DOCK LEVELLER**
VENDOR DOCUMENT NO.	:	**IPEG-6389-NTPL(1)-8470-DS-207D-0001**

Drawing/Document Category (Submission Purpose)	**Drawing/Document Review Status**			
	Approval Code		**Status**	**Action Required**
	LTHE	**OWNER/PMC**		
	1	1	Rejected	Revise & resubmit
	2	2	Commented as noted	Work may proceed subject to acceptance & incorporation of comments
⇨ For Approval	3	3	Acceptable with minor comments	Incorporate comments, if any. Resubmit as certified final. Work may proceed.
☐ For Review	4	4	Approved with No comments-FINAL	Work may proceed.
	5	5	Information Only	Accepted for information only. Work may proceed.
☐ For Information	**Owner/PMC**			
	LTHE			
		NAME	**DISCIPLINE**	**SIGN.** / **DATE**
	This approval does not relieve the Vendor / Manufacturer of the full responsibility as to design correctness of dimensions & compliance with all applicable regulations and with the requirements made in Purchaser specifications			

Rev.	**Description**	**Prepared By**	**Reviewed By**	**Approved By**	**Reviewed By**	**Approved By**	**Approved Date**
A1	For Approval	Vivek Bari	Vikrant Jadhav	Amit Kurhade	--	--	19-02-2020

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CHANGE LIST

SL. NO.	REV. NO.	PAGE NO.	CLAUSE / LINE NO.	BRIEF DESCRIPTION / REASON OF REVISION	REMARKS
1	A1	-	-	Issued for Approval	-

A Revision line in left side margin denotes revision

- DATA SHEET FOR DOCK LEVELLER -

Sr. No	Parameter	Remark
1	Required Capacity	06 Ton
2	Leveller operation mode	AC power pack hydraulically operated
3	Lip operation mode	hydraulic
4	Operation control	by push buttons and control panel
5	Loading platform height from road Level	1150 mm
6	Travel above & below dock level	600 mm
7	Platform Size (W x L)	2000 mm x 5500 mm
8	Lip size (W x L)	2000 mm x 630 mm
9	Maximum slope can be achieved	6 Degree

SIEMENS



Data sheet: Squirrel cage induction motor (Line operation)

Ordering data: 1LE7501-0DB32-3FA4

Motor type: 7CV2 803B **Frame size:** 80M **Construction:** IMB5 **Efficiency class** *IE2*

Client order no.
Order no.
Offer no.

Item no.
Consignment no.
Project

Standard Documentation

Unique Identification Code (Match this on all documents to form a set): 751_803-4_23_8778052_1-43

Rated data (Subject to tolerance as per IS:12615 / IEC 60034-1)

Phases	Voltage	Conn.	Freq.	Output	Rated Current	Rated Speed	Rated Torque		Nominal Efficiency % on sinusoidal supply			Power Factor			at % rated voltage	Locked rotor current	Locked rotor torque	Breakdown torque
	V		Hz	kW	A	rev/min			at FL	at ¾FL	at ½FL	at FL	at ¾FL	at ½FL		as multiples of rated		
																current	torque	torque
																I_A/I_N	M_A/M_N	M_K/M_N
3	415	Y	50	0.75	1.7	1433	0.51	kgf.m	79.9	79.9	77.2	0.75	0.66	0.52	100%	5.8	2.8	2.8
							5	Nm							80%	4.3	1.6	1.7

Mechanical data

Sound pressure level	60 dB(A)
Direction of rotation	Bi-directional (Standard)
Frame material	Cast Iron
Duty Type	S1
Insulation	F utilised to B
Vibration severity	A
Coating (Paint finish)	Standard Finish (Acrylic Paint)
Colour (Paint shade)	RAL 7030 - Stone Grey
Anti-condensation heaters	- / -
Motor protection	(A) - No protection (Standard)
Method of cooling	IC411
Degree of protection	IP55 (Standard)

Terminal box data

Terminal box position	TOP as viewed from DE
Material of terminal box	Cast iron
Type of terminal box	TB1L01
No. of terminals	6
Terminal thread	M6
Max. cross-sectional area	25 mm²
Cable diameter from… to…	27-35 mm
Cable entry	2xM50x1.5

Locked rotor data

Locked rotor withstand time	HOT	13s
with 50Hz, 415VY at terminals	COLD	17s

Bearings and lubrication

Bearing DE / NDE	6204 2ZC3 / 6204 2ZC3
Bearing Lifetime (L10)	50,000 hours
(for direct flexible coupling, w/o any axial load)	
Lubricant	UNIREX N3
Regreasing device	Greased for life bearings
Lubrication interval	-
Lubrication quantity	- / -
Type of bearing	

Weights and inertias

Motor weight (approx)	19 kg
Rotor weight (approx)	3.388 kg
Rotor Inertia / Rotor GD²	0.0021 kgm² / 0.0084 kgf.m²

Supply and Environmental conditions

Ambient temperature	50°C
Altitude above mean sea level	1000 m
Voltage variation	±10%
Frequency variation	±5%
Combined Voltage & Freq. variation	10%

SIEMENS

Characteristic Curves

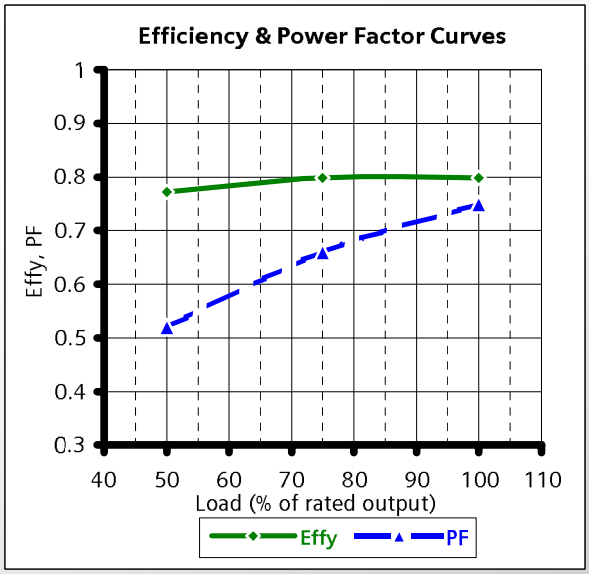
Unique Identification Code (Match this on all documents to form a set): 751_803-4_23_8778052_1-43

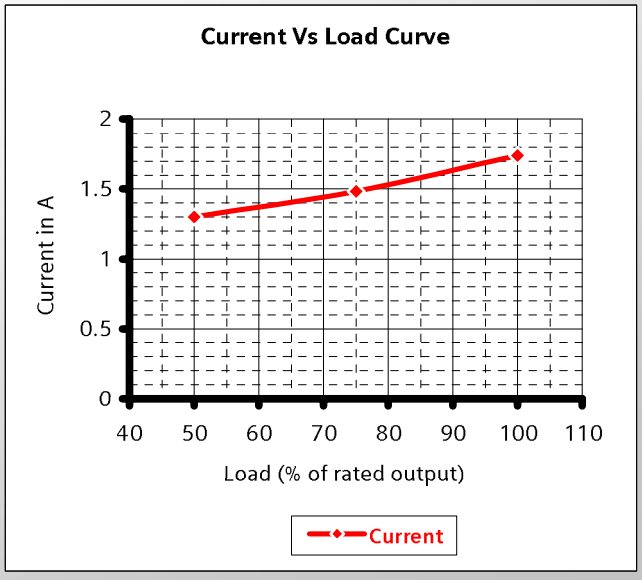
Ordering data:	1LE7501-0DB32-3FA4		Voltage : 415VY	Rated Speed :	1433 rev/min
Output :	0.75	kW	Frequency : 50Hz	Rated Torque :	0.51 kgf.m / 5 Nm
				Rated Current:	1.7A

% Load	Full Load (FL)	¾FL	½FL
Efficiency	79.9%	79.9%	77.2%
Power Factor	0.75	0.66	0.52

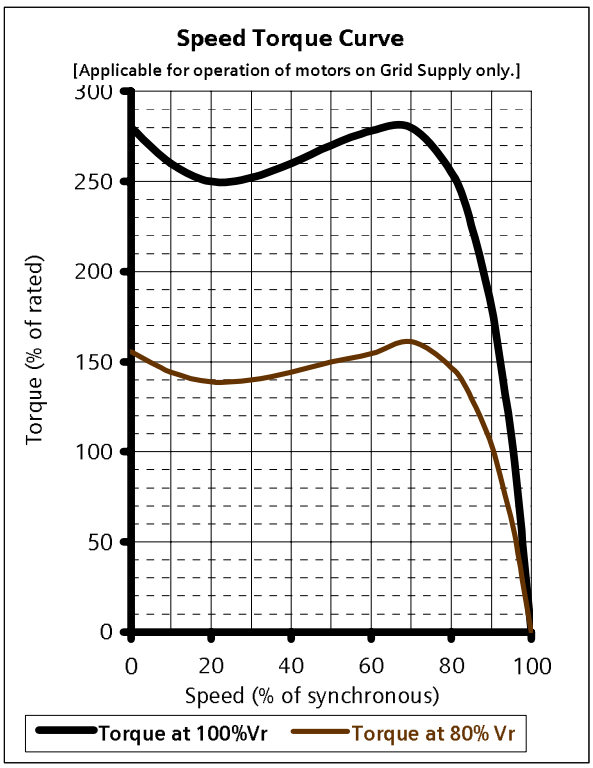
Locked Rtr Curr	5.8 x Rated
LRWT HOT	13s
LRWT COLD	17s

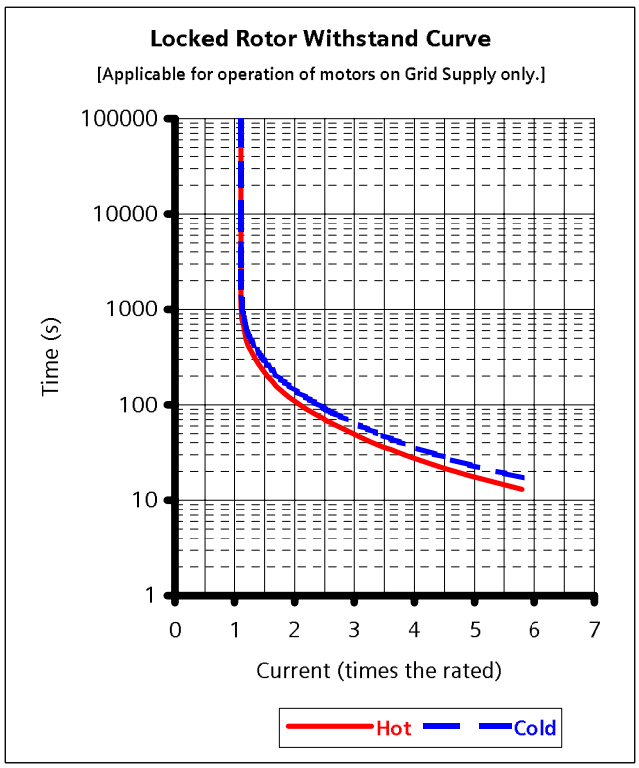
DATA SUBJECT TO TOLERANCE AS PER IS:12615 / IEC 60034-1.





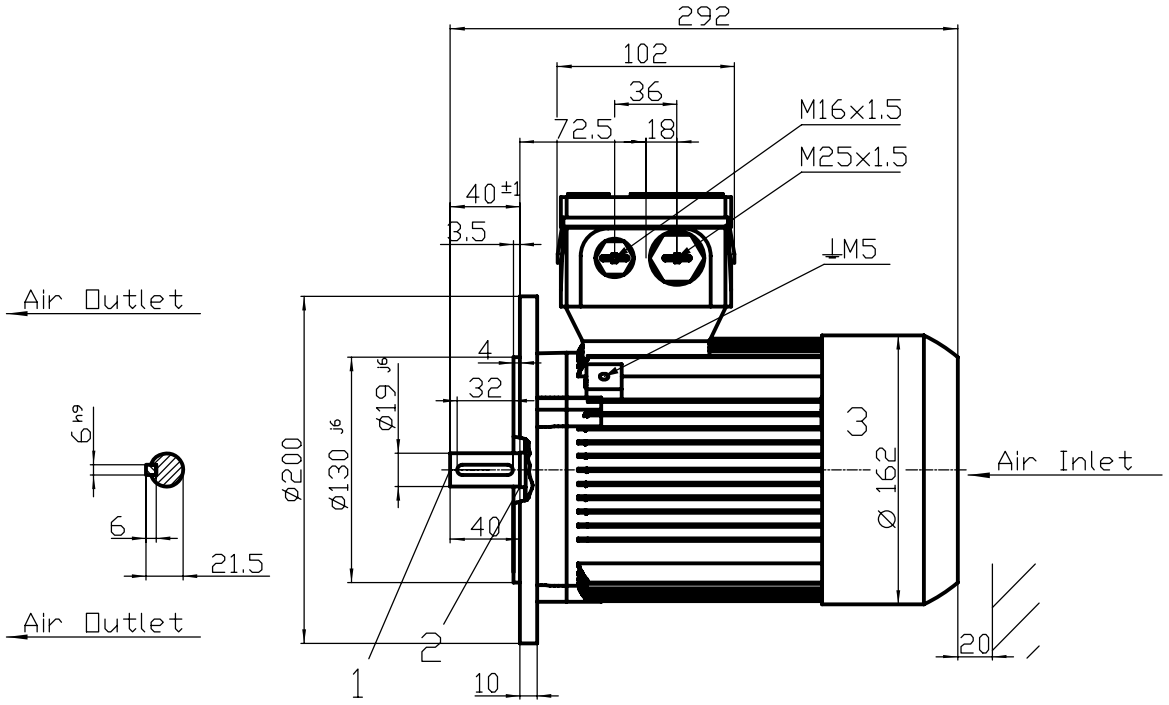
Values other than those at Full Load (FL), ¾FL and ½FL are only indicative. Do not extrapolate to No-load.

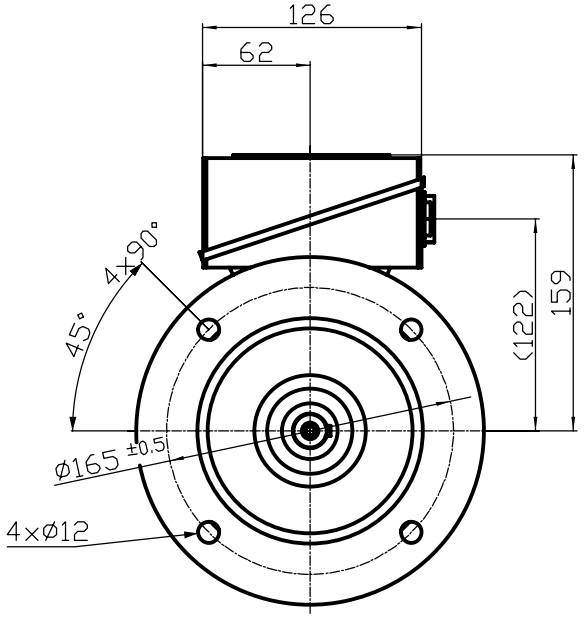




KALWA, INDIA Department : PD LD P LV-Motors	Reference Document MTR 2.6-242aa	Printed & Chkd : Atul Jugade Date : 02.02.2018 [Signature]	Format Release Date Aug '2011

Format reference: MTR 7.1-007





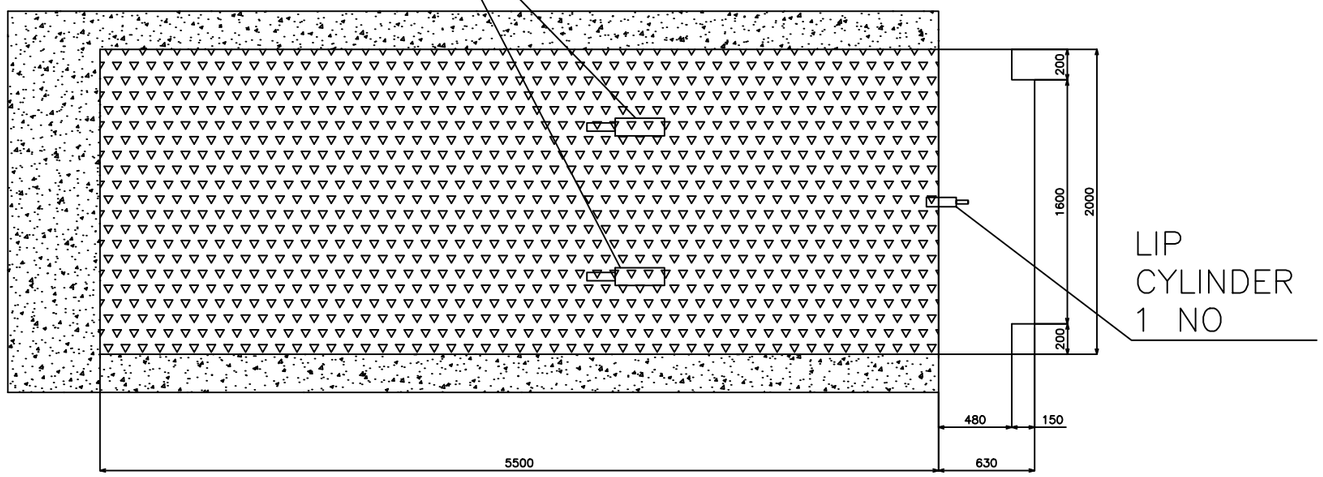
1) Protected centre hole M6x16
2) Relief groove as per DIN509 / IS:3428
3) Measured across screw heads

Tolerance: DIN EN 50347, Untolerated Dimensions- ISO 2768v / IS2102v	Surface	Material	Weight: -	Scale: mm
FS.:80M; Const.:IMB5 1LE7501-0DB32-3FA4 -	Author Creator: DTK Approval Department Change Order	Dimensional drawing		
SIEMENS	Doc State: 1/30/18 Revision Index: RS	MLFB Item No Doc No	Doc Type: / Paper Size: A4 Language: en	
© Siemens AG 2016	Project No -	Ref No -	Sheet: 1 of 1	

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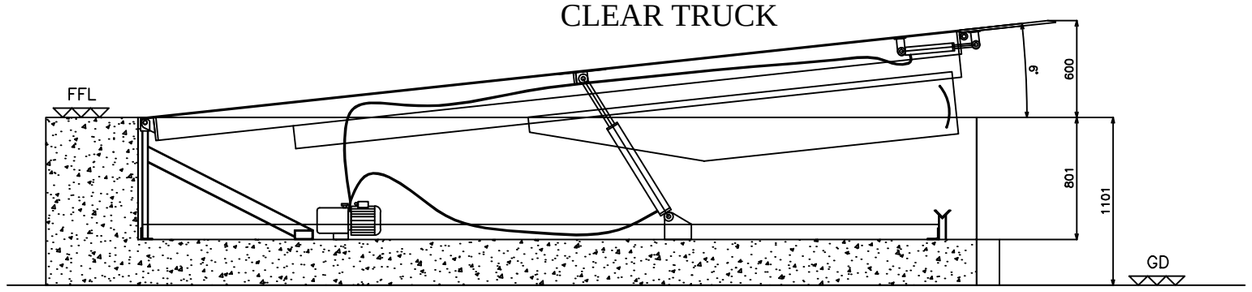
REV.	DWG NO		TITLE
XX	XXXX	1 / 1	TITLE

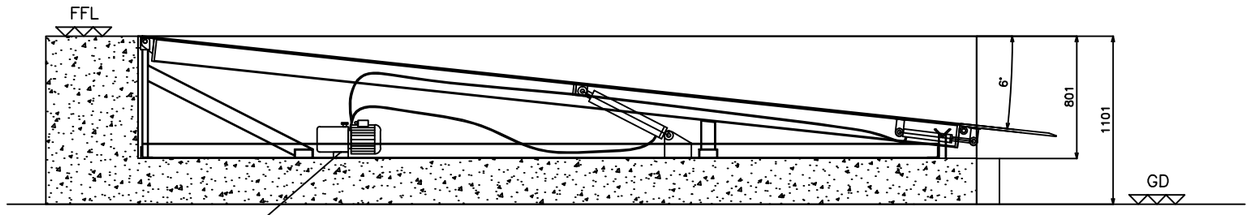
DECK CYLINDER
02 NOS



LIP
CYLINDER
1 NO

EXTRA LIFT PROVIDE TO
CLEAR TRUCK





1 HP,3 Phase HYDRAULIC
POWER PACK
(ANCHORED ON PCC FLOOR)

NOTE: –

1. PROPER DRAINAGE SHOULD BE DONE.
2. FABRICATION TOLERANCE FOR PLATFORM SIZE WILL BE + 10 MM.
3. PLEASE NOTE THAT THE CAPACITY OF DOCK LEVELER (6 TON) CONSIDERED IS THE CAPACITY AFTER THE LIP IS RESTED ON THE VEHICLE.
4. CYLINDERS PROVIDED ARE INDENTED TO BE USED ONLY FOR LIFTING OF BARE DOCK & CAN NOT LIFT OTHER LOADS.

Drawing/ Document Category (Submission Purpose)	Drawing/Document Review Status (BY OWNER/PMC)		
	APPROVAL CODE	STATUS	ACTION REQUIRED
☑ For Approval	1	Rejected	Revise and resubmit
☐ For Review	2	Commented as noted	Work may proceed subject to acceptance and incorporation of comments.
☐ For Information	3	Acceptable with minor comments	Incorporate comments, if any. Resubmit as certified final.Work may proceed.
	4	Approved with No comments–FINAL	Work may proceed.
	5	Information Only	Accepted for information only.Work may proceed.

Date: __/__/____ Name : _____

This approval does not absolve the CONTRACTOR from meeting their contractual obligations of delivering a complete safe and efficiently operating unit facilities meeting all requirements of the contract.

					/ /
					/ /
					/ /
A1	ISSUED FOR APPROVAL	PRADEEP	GAURAV	AMIT	19/02/2020
REV	PURPOSE	PREPARE	CHKD	APPROVAL	DATE dd/mm/yy

L&T - CHIYODA LIMITED

- ☐ REVIEW CODE 1 – REJECTED
  REVISE & RESUBMIT.
- ☐ REVIEW CODE 2 – COMMENTED AS NOTED. WORK MAY PROCEED SUBJECT TO ACCEPTANCE AND INCORPORATION OF COMMENTS.
- ☐ REVIEW CODE 3 – ACCEPTABLE WITH MINOR COMMENTS
  INCORPORATE COMMENTS, IF ANY. RESUBMIT AS CERTIFIED FINAL. WORK MAY PROCEED.
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  WORK MAY PROCEED.
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  ACCEPTED FOR INFORMATION ONLY. WORK MAY PROCEED.

COMMENTED BY	REVIEWED BY	APPROVED BY	DATE

THIS REVIEW / APPROVAL DOES NOT RELIEVE THE VENDOR / MANUFACTURER OF THE FULL RESPONSIBILITY AS TO DESIGN CORRECTNESS OF DIMENSIONS & COMPLIANCE WITHALL APPLICABLE REGULATIONS AND WITH THE REQUIREMENTS MADE IN PURCHASER SPECIFICATIONS.

PROJECT
ETHYLENE GLYCOL PROJECT
PARADIP REFINERY, ODISHA-INDIA.

OWNER
INDIAN OIL CORPORATION LIMITED
PARADIP REFINERY, ODISHA, INDIA

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PMC
Toyo Engineering India Private Limited
Mumbai, India

TEPL's Job No. 6389 WORK ORDER No. 25574423

LSTK–2
L&T Hydrocarbon Engineering

LSTK–2 Job No. IOCL–O&U WORK ORDER No. LATER

DETAILED ENGINEERING CENTER: **L&T - CHIYODA LIMITED**

TITLE UNIT NO–207D

DATA SHEET WITH GAD FOR
DOCK LEVELER

SCALE	1:1	DWG. NO. IPEG–6389–NTPL(1)–8470–DS–2070–0001	REV. 1/1 A1

CAD FILE REF: CFR